

Numbers

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1 Numbers

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1.2 Notes

1. The notebooks are largely self-contained, i.e, if a symbol appears, an explanation will be present at some point in the notebook.
 - Most often there will be links to the cell where the symbols are explained
 - If the symbols are not explained in this notebook, a reference to the appropriate notebook will be provided
2. **Github does a poor job of rendering this notebook.** The online render of this notebook is missing links, symbols, and notations are badly formatted. It is advised to clone a local copy (or download the notebook) and open it locally.

3. See the Collections notebook before this notebook to gain familiarity with set notations

1.3 Importing Libraries

```
[1]: import random
import math
import cmath
```

Real numbers

Absolute value

The absolute value or modulus of a real number x is the non-negative value of x without regard to its sign, and it is denoted by: $|x|$

$$|x| = \begin{cases} x, & \text{if } x \geq 0 \\ -x, & \text{if } x < 0 \end{cases}$$

For more information on the piecewise conditional braces see the Functions notebook

```
[2]: abs(2), abs(-2)
```

```
[2]: (2, 2)
```

Plus minus

The plus minus notation is used to encapsulate the positive and negative value of a real number together as a shorthand. It may be denoted for a real number x using:

$$\pm x$$

or

$$\mp x$$

Both notations mean the same. But the second notation may be called **minus plus** notation. For example the solutions to the equation $x^2 - 4 = 0$ is 2 and -2 , which may be written together as ± 2

```
[3]: # x^2 - 4 = 0 has solutions x = ±2
x = 2

print(f'(+{x})^2 - 4 = {x**2 - 4}')
print(f'(-{x})^2 - 4 = {(-x)**2 - 4}')
```

```
(+2)^2 - 4 = 0
```

```
(-2)^2 - 4 = 0
```

When the *plus minus* symbol is repeated, it can make the meaning ambiguous, for example:

$$\pm 1 \pm \sqrt{3}$$

may mean:

$$1 + \sqrt{3}, 1 - \sqrt{3}, -1 + \sqrt{3}, -1 - \sqrt{3}$$

But when the *minus plus* symbol is used with the *plus minus* notation, the meaning may be quite clear:

$$\mp 1 \pm \sqrt{3}$$

may mean:

$$-1 + \sqrt{3}, 1 - \sqrt{3}$$

Context can be important in these cases

Defined

The defined symbol is used to show the definition of a variable. For example, a variable x defined as $1 - \sqrt{2}$ is denoted as:

$$x := 1 - \sqrt{2}$$

or

$$x \stackrel{\text{def}}{=} 1 - \sqrt{2}$$

or

$$x \triangleq 1 - \sqrt{2}$$

```
[4]: x = 1 - math.sqrt(2)
```

```
x
```

```
[4]: -0.41421356237309515
```

Some authors use the $:=$ symbol for iterative definition. Let i be the current iteration, then the iterative rule can be written as

$$x := x + 1$$

which is traditionally written as

$$x_{i+1} = x_i + 1$$

```
[5]: # x := x + 1 applied 10 times starting from x = 10
```

```
x = 10
```

```
for _ in range(10):
```

```
    x = x + 1
```

```
x
```

[5]: 20

Identically equal to

The identically equal to symbol can be used to show that an assertion is equal for **all** values, and is denoted by: $\equiv$

For example:

$$x \in \mathbb{R}, \sin^2 x + \cos^2 x \equiv 1$$

is true for **all** values of x . Compare this with the assertion $x^2 - 4 = 0$ which is true only for *some* values of x .

(See: Real set)

```
[6]: # sin^2(x) + cos^2(x) = 1 for all x
for _ in range(5):
    x = random.random()
    result = math.sin(x) ** 2 + math.cos(x) ** 2
    print(f'x = {x:.4f}: sin^2(x) + cos^2(x) = {result}')
```

```
x = 0.9513: sin^2(x) + cos^2(x) = 1.0
x = 0.4616: sin^2(x) + cos^2(x) = 1.0
x = 0.1651: sin^2(x) + cos^2(x) = 1.0
x = 0.9748: sin^2(x) + cos^2(x) = 1.0000000000000002
x = 0.6947: sin^2(x) + cos^2(x) = 1.0000000000000002
```

```
[7]: # x^2 - 4 = 0 is true only for specific values of x (not an identity)
for x in [2, -2, 3]:
    print(f'x = {x:2}: x^2 - 4 = {x**2 - 4:2}, equals 0: {x**2 - 4 == 0}')
```

```
x =  2: x^2 - 4 =  0, equals 0: True
x = -2: x^2 - 4 =  0, equals 0: True
x =  3: x^2 - 4 =  5, equals 0: False
```

Subset of the Reals

Real set

A set containing real numbers is called a real number set. It is represented by:

$$\mathbb{R}$$

Membership of variable x in the real number set is represented as:

$$x \in \mathbb{R}$$

```
[8]: x = 3.8
```

```
# x R: check if x is a real number
isinstance(x, (float, int))
```

[8]: True

Integer set

A set containing only integers is called an integer set. It is represented by:

$$\mathbb{Z}$$

Membership of variable x in the integer set is represented as:

$$x \in \mathbb{Z}$$

```
[9]: x = 5

# x Z: check if x is an integer
isinstance(x, int)
```

[9]: True

Natural number set

The definition of natural number is not standard, but the definition that is preferred is that a natural number is a non-negative integer. It is denoted by:

$$\mathbb{N} = \{0, 1, 2, \dots\}$$

However, some authors and standards prefer not to include 0 such that $\mathbb{N} = \{1, 2, \dots\}$

```
[10]: x = 5

# x N: check if x is a non-negative integer
isinstance(x, int) and x >= 0
```

[10]: True

Rational number set

Rational numbers are those numbers that can be expressed as a ratio of integers $\frac{a}{b}$ where $b \neq 0$ and is denoted by: $\mathbb{Q}$

$$\mathbb{Q} = \{x \in \mathbb{R} \mid x = a/b \text{ where } a, b \in \mathbb{Z} \text{ and } b \neq 0\}$$

```
[11]: # Q = {a/b : a, b ∈ Z, b ≠ 0}
examples = [(1, 2), (3, 7), (-5, 3), (22, 7)]

for a, b in examples:
    print(f'a = {a:3}, b = {b}: a/b = {a/b:.6f}')
```

```
a = 1, b = 2: a/b = 0.500000
a = 3, b = 7: a/b = 0.428571
a = -5, b = 3: a/b = -1.666667
a = 22, b = 7: a/b = 3.142857
```

These sets are nested as follows:

$$\mathbb{R} \supset \mathbb{Q} \supset \mathbb{Z} \supset \mathbb{N}$$

Higher dimensional spaces

The n -dimensional number coordinate space is the set of all ordered n -tuples of the respective numbers. This is represented with a superscript over the number set notation. For example, the real coordinate space of dimension n is denoted as:

$$\mathbb{R}^n$$

where $\mathbb{R}^n = \{(a_1, a_2, a_3, \dots, a_n) \mid a_i \in \mathbb{R}\}$

And the n -dimensional complex coordinate space (or complex n -space) is denoted as:

$$\mathbb{C}^n$$

where $\mathbb{C}^n = \{(c_1, c_2, c_3, \dots, c_n) \mid c_i \in \mathbb{C}\}$

Additional commonly seen example: Points within a 3-dimensional Euclidean space is considered to form a real coordinate space of dimension three: $\mathbb{R}^3$, such that if $\mathbf{x} \in \mathbb{R}^3$, $\mathbf{x}$ may look like $\mathbf{x} = \{(x_1, x_2, x_3)_1, (x_1, x_2, x_3)_2, \dots\}$

For a Vectors perspective of n dimensional real vectors please see the Linear Algebra notebook

```
[12]: # points in R^3
p_1 = (1.0, -2.5, 3.7)
p_2 = (0.0, 4.1, -1.3)

print(f'p_1 = {p_1}, dimension: {len(p_1)}')
print(f'p_2 = {p_2}, dimension: {len(p_2)}')
```

```
p_1 = (1.0, -2.5, 3.7), dimension: 3
p_2 = (0.0, 4.1, -1.3), dimension: 3
```

Decorations

The number-set notations can be decorated with additional symbols. Some of these decorations are standard, while others vary by author and by context.

Nonnegative, positive, nonpositive, and negative parts of a set are written most clearly with subscripts:

-

$$\mathbb{Z}_{\geq 0}, \mathbb{Z}_{>0}, \mathbb{Z}_{\leq 0}, \mathbb{Z}_{<0}$$

-

$$\mathbb{R}_{\geq 0}, \mathbb{R}_{>0}, \mathbb{R}_{\leq 0}, \mathbb{R}_{<0}$$

(See: Natural number set)

Superscripts such as $\mathbb{Z}^+$, $\mathbb{Z}^-$, $\mathbb{R}^+$, and $\mathbb{R}^-$ are also used, but their meanings are not completely uniform across books.

For star decorations, context is especially important:

- $\mathbb{R}^*$ usually denotes the nonzero real numbers:

$$\mathbb{R}^* = \{x \in \mathbb{R} \mid x \neq 0\}$$

- $\mathbb{C}^*$ usually denotes the nonzero complex numbers:

$$\mathbb{C}^* = \{z \in \mathbb{C} \mid z \neq 0\}$$

- $\mathbb{Z}^\times = \{-1, 1\}$ denotes the units of $\mathbb{Z}$
- The extended real line is more clearly written as $\overline{\mathbb{R}}$, and the Riemann sphere is more clearly written as $\hat{\mathbb{C}}$ or $\overline{\mathbb{C}}$

Note: Decorations are often author-specific, so surrounding context should always be checked.

Cardinality of ints and reals

The cardinality of sets is represented using two pipes, for example: $|A|$ (*See Collections notebook for more info*). But there are special symbols to represent cardinality of integers and reals.

The cardinality of integers is denoted using: $\aleph_0$

$$\aleph_0 = |\mathbb{Z}|$$

$\aleph_0$ (read as aleph null) is the smallest transfinite cardinal number and sets with cardinality $\aleph_0$ are called countable.

The cardinality of real numbers is denoted using: c

$$c = |\mathbb{R}|$$

c is also called the cardinality of the continuum.

Intervals

Intervals indicate membership of elements within a range and are denoted as open or closed using paranthesis or brackets respectively. A paranthesis indicates endpoints that are not included, a bracket indicates an endpoint that is included. For example $[-1,1]$ indicates all real numbers beginning from -1 inclusive and ending at 1 inclusive. Other examples:

$$[1,2] = \{x \in \mathbb{R} \mid 1 \leq x \leq 2\}$$

$$[1,2) = \{x \in \mathbb{R} \mid 1 \leq x < 2\}$$

$$(1,2] = \{x \in \mathbb{R} \mid 1 < x \leq 2\}$$

$$(1,2) = \{x \in \mathbb{R} \mid 1 < x < 2\}$$

```
[13]: #[1,2]

for x in [0.5, 1, 1.5, 2, 2.5]:
    print('\nx: ',x)
    print('[1,2]: ', 1<=x<=2)
    print('[1,2): ', 1<=x<2)
    print('(1,2]: ', 1<x<=2)
    print('(1,2): ', 1<x<2)
```

```
x: 0.5
[1,2]: False
[1,2): False
(1,2]: False
(1,2): False
```

```
x: 1
[1,2]: True
[1,2): True
(1,2]: False
(1,2): False
```

```
x: 1.5
[1,2]: True
[1,2): True
(1,2]: True
(1,2): True
```

```
x: 2
[1,2]: True
[1,2): False
(1,2]: True
(1,2): False
```



```
x: 2.5
[1,2]: False
[1,2): False
(1,2]: False
(1,2): False
```

The symbols $-\infty$ and ∞ may be used for the left and right ends to show unbounded ranges:

$$\begin{aligned}[2, \infty) &= \{x \in \mathbb{R} \mid x \geq 2\} \\ (-\infty, -2) &= \{x \in \mathbb{R} \mid x < -2\} \\ (-\infty, \infty) &= \mathbb{R}\end{aligned}$$

Some authors use reversed brackets to denote exclusive endpoints, it is the same as using a parenthesis:

$$\begin{aligned}[a, b[&\text{is equivalent to } [a, b) \\]a, b[&\text{is equivalent to } (a, b)\end{aligned}$$

Famous numbers

Some constants have their own symbols and used generally enough to be mentioned here.

- Pi is the ratio a circle's circumference to it's diameter and denoted using: π
- Euler's number or the base of natural logarithms is denoted using: e
- The imaginary unit with it's value equal to $\sqrt{-1}$ is denoted by: i
- Physical constants may include speed of light (denoted by c) etc

More info: https://en.wikipedia.org/wiki/Mathematical_constant

Complex numbers

Complex set

A set containing only complex numbers is called a complex number set. It is represented by:

$$\mathbb{C}$$

Membership of variable z in the complex number set is represented as:

$$z \in \mathbb{C}$$

```
[14]: z_1 = complex(3, 4)
      z_2 = 3 + 6j
```

```
print(f'z_1 = {z_1}')
print(f'z_2 = {z_2}')
isinstance(z_1, complex), isinstance(z_2, complex)
```

```
z_1 = (3+4j)
z_2 = (3+6j)
```

```
[14]: (True, True)
```

The complex numbers are created using an object i which has the property $i^2 = -1$ sometimes called the imaginary unit. The result is a collection of numbers of the form:

$$\{a + ib \mid a, b \in \mathbb{R}, i^2 = -1\}$$

Some authors prefer writing i after b for ex, $a + bi$, and in electrical engineering literature authors prefer to use j instead of i since i represents electrical current: $a + bj$

Here, a is called the real component and b is called the imaginary component, and it is generally denoted using: $\Re$ for real, and $\Im$ for imaginary.

$$z = a + bi, \Re z = a, \Im z = b$$

This can also be represented using simply the abbreviations as:

$$z = a + bi, \operatorname{Re} z = a, \operatorname{Im} z = b$$

```
[15]: z = complex(3,4)
      z.real, z.imag
```

```
[15]: (3.0, 4.0)
```

```
[16]: z = 3 + 6j
      z.real, z.imag
```

```
[16]: (3.0, 6.0)
```

Polar representation

The complex number $z = a + bi$ can simply be represented in a complex plane as points with coordinates (a, b) . These can also be represented in polar coordinates (r, θ) where simple rectangular to polar transformation rules apply:

$$a = r \cos \theta, b = r \sin \theta$$

Extended to complex numbers, the polar representation of $z = a + ib$ can be denoted as: $re^{i\theta}$ or $r \operatorname{cis} \theta$

$$re^{i\theta} = r[\cos \theta + i \sin \theta] = a + ib$$

or

$$r \operatorname{cis} \theta = r[\cos \theta + i \sin \theta] = a + ib$$

Here, r is the magnitude (See: Magnitude/Absolute value), and θ is the argument (or also called phase angle) and is denoted by $\arg(z)$

$$r = |z| = |a + ib|$$

$$\theta = \arg(z) = \arg(a + ib)$$

```
[17]: z = complex(3,4)
      r = abs(z)
      theta = cmath.phase(z)

      r, theta
```

```
[17]: (5.0, 0.9272952180016122)
```

Magnitude/Absolute value

The magnitude or absolute value of a complex number z is denoted as: $|z|$

$$z = a + bi, |z| = |a + bi| = \sqrt{a^2 + b^2}$$

```
[18]: z = complex(3,4)

      abs(z)
```

```
[18]: 5.0
```

Conjugate

The (complex) conjugate of a complex number z is denoted as: $\bar{z}$

$$z = a + bi, \bar{z} = \overline{a + bi} = a - bi$$

```
[19]: z = complex(3,4)

      z.conjugate()
```

```
[19]: (3-4j)
```